

We recognize and acknowledge that McMaster University meets and learns on the traditional territories of the Mississauga and Haudenosaunee nations, and within the lands protected by the "[Dish With One Spoon](#)" wampum, an agreement amongst all allied Nations to peaceably share and care for the resources around the Great Lakes.

MATH 2Z03 – Engineering Mathematics III

2024 Fall Term

Instructor: Dr N Kevlahan | **E-mail:** math2z03@mcmaster.ca | **Office:** HH-324 | **Office Hour:** We 16:00-17:00

Course website: [Avenue to Learn](#)

Course Description

Ordinary differential equations, introduction to partial differential equations, applications to problems from engineering.

Differential equations are essential for modelling many practical problems arising in engineering such as heat transfer, aeronautics, structural engineering, and electro-magnetism.

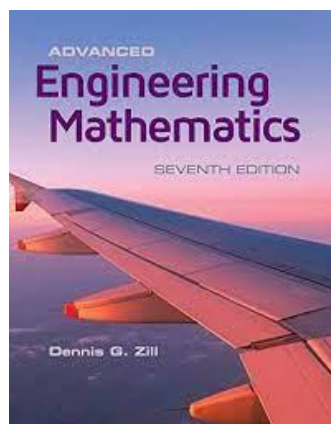
You will learn how to solve first- and second-order linear ordinary and partial differential equations using separation of variables, the linear superposition principle, and the Laplace transform. We primarily focus on analytical solutions but will also do some computational modelling in the labs.

Prerequisites: MATH 1ZB3 or 1ZC3

Antirequisite: ENGINEER 2Z03, MATH 2C03, 2M03, 2P04

Materials & Fees

Textbooks



Advanced Engineering
Mathematics

ISBN: 978-1-284-25008-4

Authors: Dennis G. Zill

Publisher: Jones and Bartlett
Learning

Publication Date: 2022

Edition: 7th

Software

We will use [childsmath](#) for the assignments, the computer labs and the two mid-term tests.

The computer labs and associated tutorials use [matlab](#), which is free for all students.



In-person Delivery

This course **will run fully in-person**.

Echo360 recordings will be available on the Avenue course page a few hours after the lecture.

I will post lecture notes in pdf format on Avenue.

Course Evaluation

There will be 6 online assignments, 5 online labs, 2 online mid-term tests, and an in-person final exam.

To foster an inclusive learning environment and reduce the administrative burden on students seeking accommodations, **in consultation with Student Accommodation Services (SAS)**, a series of pre-emptive accommodations have been activated.

These universal measures address the most common barriers to learning experienced by MATH 2Z03 students. **There is no need to file an accommodation request or to contact the instructor to receive the accommodations outlined as ‘universal design’ below** – you are already approved and have access to these features, extensions, and modifications.

If you seek additional accommodations that are not covered by the preapproved universal design list – for example, a braille or large print copy of the textbook; writing tests/exams at the testing centre – you can engage in the student accommodation process to receive these accommodations as outlined below in the ‘Academic Accommodations for Students with Disabilities’ and ‘Academic Accommodations for Religious, Indigenous Or Spiritual Observations (RISO)’ sections.

Assignments

The assignments are completed online using [childsmath](#):

Assignment	Assignment due
Assignment 1	Thursday September 19 at 23:59
Assignment 2	Thursday October 3 at 23:59
Assignment 3	Thursday October 17 at 23:59
Assignment 4	Thursday November 7 at 23:59
Assignment 5	Thursday November 21 at 23:59
Assignment 6	Thursday December 5 at 23:59

You are allowed **3 attempts on each question** and **5 attempts on each assignment**.

To support **universal design**, although the assignments are due on the Thursday at 23:59, all students are granted a **72-hour grace period**: you may submit the assignment by the **following Sunday at 23:59** with no penalty.



Assignment solutions will be discussed in the tutorials immediately after the extended grace period deadline. This means that **assignments submitted after the grace period cannot be accepted**.

SAS has confirmed that this universal **design automatically satisfies** “Consideration for extension on assignments/lab reports up to one week” **accommodations**. You do not need to contact the instructor regarding extra time accommodations.

Computer laboratories

The labs are completed online using [childsmath](#):

Laboratory	Laboratory due
Lab 1	Thursday September 26 at 23:59
Lab 2	Thursday October 10 at 23:59
Lab 3	Thursday October 31 at 23:59
Lab 4	Thursday November 14 at 23:59
Lab 5	Thursday November 28 at 23:59

Most labs have associated tutorials available on [childsmath](#) that you should review before attempting the lab. The labs require matlab. You are **allowed 3 attempts on each question** and **5 attempts on each lab**.

Although the assignments are due on the Thursday at 23:59, to support **universal design**, all students are granted a **72-hour grace period**. You may submit the assignment by the **following Sunday at 23:59** with no penalty.

SAS has confirmed that this universal design **automatically satisfies** “Consideration for extension on assignments/lab reports up to one week” **accommodations**. You do not need to contact the instructor regarding extra time accommodations.

Tests

There will be **two mid-term tests**:

- Mid-term 1: available **Monday October 21 at 08:00 to Monday October 28 at 08:00**.
- Mid-term 2: available **Monday November 18 at 08:00 to Monday November 25 at 08:00**.

The mid-term tests will have multiple choice and true/false questions. They will be held online using [childsmath](#). Only one attempt is allowed on each test.

The tests **are designed to take one hour**, but to support **universal design** all students **will have two hours to complete them**. You may attempt the tests in **any 2-hour window during the weeks** indicated above.

SAS has confirmed that this universal test design **normally satisfies** the following accommodations:



- "Extra time to complete written/ multiple choice tests" (*up to 60 minutes/hr additional time*)
- "Student allowed one full day of rest between scheduled tests."
- "Student allowed to take breaks to stretch and move about during tests."
- "Student allowed to write ONLY one test per day"
- "Able to write tests/exams in the [morning/afternoon/evening]"
- "Use of a computer for all tests"
- "Student may write tests/exams in a small group room (or alone)"

There is no need to contact the instructor or SAS regarding these accommodations.

Final exam

There will be a **2.5-hour final exam**, scheduled by the registrar.

Grading System

The final grade will be calculated using the maximum of Grading Schemes I and II as follows:

Assessment	Grading scheme I	Grading Scheme II
Assignments	15% (<i>best 5 of 6</i>)	15% (<i>best 5 of 6</i>)
Computer labs	15% (<i>best 4 of 5</i>)	15% (<i>best 4 of 5</i>)
Two mid-term tests	30% (<i>15% each</i>)	20% (<i>best 1 of 2</i>)
Final Exam	40%	50%

Important note: If you **miss one mid-term test** then your grade will automatically be calculated using Grading scheme 2. If you **miss both mid-term tests** you need to contact the instructor and you will be given an individualized **oral exam**.

In addition, for either grading scheme, **if the mid-term mark is lower than the final exam mark**, then it will be replaced by the final exam mark to boost your grade (e.g. 55% or 70% final).

Instructor Policy for Requests for Relief for Missed Academic Term Work (MSAFs)

If you **MSAF an assignment**, that assignment will be treated as the lowest mark assignment and your assignment mark will be based on the other five assignments (as per the grading system).

If you **MSAF a lab**, that lab will be treated as the lowest mark lab and your lab mark will be based on the other four labs (as per the grading system).

If you **MSAF a test**, that test will be treated as your lowest test mark and Grading Scheme II will be used (as per the grading system).

Bottom line: you should not need to use your MSAF as it will not affect the evaluation scheme and cannot improve your final grade or provide more flexibility than the standard grading schemes!

Lecture schedule (*tentative*)

Section numbers refer to the textbook (Zill, 7th edition).

Lecture	Topic
Week 1	
1	1.1 Definitions and terminology (<i>first class</i>)
2	1.2 Initial value problems
Week 2	
3	2.1 Direction fields and autonomous equations
4	(2.1 continued)
5	2.2 Separable equations
Week 3	
6	2.3 Linear equations
7	(2.3 continued)
8	2.7 Linear models (<i>exponential growth and decay, Newton's law of cooling</i>)
Week 4	
9	3.1 Linear second-order equations
10	(3.1 continued)
11	3.3 Homogeneous linear equations with constant coefficients
Week 5	
12	(3.3 continued)
13	3.4 Method of undetermined coefficients
14	(3.4 continued)
Week 6	
15	3.5 Method of variation of parameters
16	(3.5 continued)
17	3.6 Cauchy-Euler equations
Week 7 (test 1 week)	
18	3.8 Initial value problems for linear equations (<i>oscillators and damping</i>)
19	(3.8 continued)
20	3.9 Boundary value problems for linear equations (<i>beams and buckling</i>)
Week 8	
21	4.1 Definition of the Laplace transform
22	(4.1 continued)
23	4.2 Inverse Laplace transform, and Laplace transform of derivatives
Week 9	
24	4.3 Translation of Laplace transform



25	(4.3 continued, including unit step function)
26	4.4 Additional properties of Laplace transform
Week 10	
27	(4.4 continued, including convolutions)
28	4.5 The Dirac delta function
29	13.1 Separable partial differential equations
Week 11 (test 2 week)	
30	(13.1 continued)
31	13.2 Classical boundary value problems for partial differential equations
32	(13.2 continued)
Week 12	
33	13.3 Heat equation
34	(13.3 continued)
35	13.4 Wave equation
Week 13	
36	(13.4 continued)
37	Catch up and review (last class)

Important message

The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of students to check their McMaster email and course websites weekly during the term and to note any changes. Announcements will be made in class and by using the course email distribution list.

Policy Regarding Missed Work

Requests for missed academic work worth less than 25% of the final grade resulting from personal or medical situations, lasting up to 3 calendar days, can be reported, once per term, without documentation, using the McMaster Student Absence Form (MSAF). Relief for missed work for a longer duration or for other reasons must be reported to your Faculty office, and relief from term work may not necessarily be granted. When using the MSAF, report your absence to the course instructor or designate. You must then contact the Instructor/Instructional Assistant/other immediately (normally within 2 working days) by e-mail.

Please refer to the contact list on the first page of this outline for appropriate e-mail addresses. The Instructor/Instructional assistant will indicate what relief may be granted for the work you have missed, and relevant details such as revised deadlines, or time and location of a make-up exam/quiz/test. Please note that the MSAF may not be used for final deliverables, nor can it be used for a final examination or its equivalent. Please [review and follow the Academic Regulation in the Undergraduate Calendar under "Requests for Relief for Missed Academic Term Work" here.](#)



Academic Accommodation of Students with Disabilities

Students with disabilities who require academic accommodation must contact [Student Accessibility Services \(SAS\)](#) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's [Academic Accommodation of Students with Disabilities](#) policy.

Academic Accommodation for Religious, Indigenous Or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office **normally within 10 working days** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Courses with an On-Line Element

This course will use Avenue to Learn and Echo360. Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, usernames for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure, please discuss this with the course instructor.

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity.

It is your responsibility to understand what constitutes academic dishonesty.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy, located here.](#)

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Use of Artificial Intelligence ("AI") is Prohibited

Students are not permitted to use generative AI in this course. In alignment with [McMaster academic integrity policy](#), it "shall be an offence knowingly to ... submit academic work for assessment that was purchased or acquired from another source". This includes work created by generative AI tools. Also state in the policy is the following, "Contract Cheating is the act of "outsourcing of student work to third

parties” (Lancaster & Clarke, 2016, p. 639) with or without payment.” Using Generative AI tools is a form of contract cheating. Charges of academic dishonesty will be brought forward to the Office of Academic Integrity.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities \(the “Code”\)](#). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online**.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

Senate Student Policies

[Students can view full policies here on the University Secretariat website.](#) Senate Policy Statements are also available from the Senate Secretariat Office, Room 104, and Gilmour Hall.

Student Code of Conduct

You acknowledge that your behavior in all aspects of this course should meet the standards of the McMaster University Student Code of Conduct. You understand that any inappropriate behavior directed against any of your colleagues, teaching assistants, or the instructional team will not be tolerated. Disruptive behavior during any session (e.g. lecture, seminar, lab, tutorial) such as talking, sleeping or non-class computing while an individual presents information, or constantly being late, will also not be tolerated. Abuse, ridicule, slander, inappropriate language, and discrimination towards instructors teaching staff, teaching assistants and other students will not be tolerated in any capacity. Shared spaces including e-spaces such as the Avenue to Learn course discussion board are to be considered inclusive and safe. [For more information, please follow this link.](#)

Inclusivity and Academic Integrity

The University values integrity, inclusiveness, and teamwork, and strives to support the personal and collective growth of the McMaster student community. These values are foundational to ensuring campus environments – both in-person and virtual – are conducive to personal wellbeing and academic success.

Inclusive Learning

McMaster University aims to foster a supportive and inclusive learning environment that encourages both individual and collective growth. Students are invited to speak with the Instructor immediately if they

encounter any challenges related to accessing or using the technological requirements expected for successful participation in this course.

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors. The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Additional Copyright Information

In this course students will have access to material that is subject to copyright laws. This includes (but is not limited to) textbooks and all resources developed by the Instructor such as lab manuals, demonstration videos, quizzes, assignments, tests, class notes and class slides. Students are not allowed to share or redistribute this material in any printed or electronic form without the explicit written consent of the copyright holder. This includes posting any course material on Internet bulletin boards, course repositories, social networks, etc.

Research Ethics

NA

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, A2L and/or McMaster email.